

Introduction to Real Analysis – Homework 5

The unsigned Lebesgue integral

Notation. m is Lebesgue measure, $\mathbf{1}_A$ is the indicator of A , and all integrals are Lebesgue integrals. Use $0 \cdot \infty = 0$.

1. Countably many levels.

- (a) Let $A_1, A_2, \dots \subset \mathbb{R}^d$ be pairwise disjoint measurable sets and let $a_n \geq 0$ be real numbers. Prove that $f = \sum_{n \geq 1} a_n \mathbf{1}_{A_n}$ is measurable and

$$\int_{\mathbb{R}^d} f \, dm = \sum_{n \geq 1} a_n m(A_n).$$

- (b) On $\mathbb{R}$, compute $\int_{\mathbb{R}} h \, dm$ for

$$h(x) = \sum_{n \geq 1} n \mathbf{1}_{(2^{-n}, 2^{1-n}]}(x).$$

2. Increasing domains and tails.

Let $f : \mathbb{R}^d \rightarrow [0, \infty]$ be measurable.

- (a) If measurable sets E_n increase to E , prove that

$$\lim_{n \rightarrow \infty} \int_{E_n} f \, dm = \int_E f \, dm.$$

Hint: test against simple functions below $f \mathbf{1}_E$.

- (b) If $\int_{\mathbb{R}^d} f \, dm < \infty$, deduce that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^d \setminus [-n, n]^d} f \, dm = 0.$$

3. Uniform convergence.

Let $E \subset \mathbb{R}^d$ be measurable with $m(E) < \infty$. Let f_n, f be bounded nonnegative measurable functions, all zero outside E .

- (a) If $f_n \rightarrow f$ uniformly, prove that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^d} f_n \, dm = \int_{\mathbb{R}^d} f \, dm.$$

- (b) Show that this conclusion can fail when $m(E) = \infty$.

4. A pointwise limit.

Construct nonnegative simple functions $u_n : \mathbb{R} \rightarrow [0, \infty)$, all zero outside $[0, 1]$, such that $u_n(x) \rightarrow 0$ for every x , but

$$\int_{\mathbb{R}} u_n \, dm = 1 \quad \text{for every } n.$$

Explain why this does not contradict the theorem on increasing simple approximations.